

Estimating a clustered localization radius for the modified-Cholesky ensemble Kalman filter: a combinatorial formulation

Working note — not a submission draft

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Abstract

This note sets out, in one place, what the experiment suite computes and why it is built the way it is. The localization radius of a modified-Cholesky ensemble Kalman filter is treated as a vector of integers, one per cluster of grid points, where the clusters come from ensemble-derived features rather than from position. The resulting problem is combinatorial with $r_{\max}^{2K}$ configurations, and six metaheuristics are compared on it against two controls. The quantity being minimized is computed from observations alone, by withholding part of the observation lattice; an oracle that uses the true state is run alongside as an upper bound on what any selection rule could achieve. Several design decisions that look arbitrary are consequences of measurements reported here, and two open problems are stated rather than hidden.

1 What the problem is

The analysis step of an ensemble Kalman filter combines a forecast ensemble with observations. With an ensemble far smaller than the state dimension, the sample covariance is rank deficient and carries spurious correlations between distant variables. Localization removes them, and every localization scheme depends on a radius of influence.

That radius is normally chosen by sweeping it against a known truth in a synthetic experiment and then freezing it. The question here is whether it can be *estimated* instead — from observations alone — and whether it pays to let it vary in space.

1.1 The filter

We use the modified-Cholesky estimator of the precision matrix [1, 2]. Let $\mathbf{X}^b \in \mathbb{R}^{n \times N}$ be the forecast ensemble, $\Delta \mathbf{X}$ its deviations from the mean. For each component i with predecessor set $P(i, r_i)$ — the neighbours of i within radius r_i whose index is lower — solve the ridge regression

$$\beta_i = (\mathbf{X}_P^\top \mathbf{X}_P + \alpha \mathbf{I})^{-1} \mathbf{X}_P^\top \Delta \mathbf{X}_i, \quad \mathbf{X}_P = \Delta \mathbf{X}_{P(i, r_i)}^\top, \quad (1)$$

and assemble

$$\mathbf{L}_{ii} = 1, \quad \mathbf{L}_{i, P(i, r_i)} = -\beta_i^\top, \quad \mathbf{D}_{ii} = \text{var}(\Delta \mathbf{X}_i - \mathbf{X}_P \beta_i)^{-1}, \quad \hat{\mathbf{B}}^{-1}(\mathbf{r}) = \mathbf{L}^\top \mathbf{D} \mathbf{L}. \quad (2)$$

The analysis then solves $(\hat{\mathbf{B}}^{-1} + \mathbf{H}^\top \mathbf{R}^{-1} \mathbf{H}) \delta \mathbf{x} = \mathbf{H}^\top \mathbf{R}^{-1} (\mathbf{y} - \mathbf{H} \mathbf{x}^b)$.

Two properties of (2) drive everything that follows.

Remark 1 (Row separability). *Row i of $\mathbf{L}$ and entry $\mathbf{D}_{ii}$ depend on r_i and on nothing else. Changing one component of the radius vector therefore leaves every other row untouched.*

Remark 2 (Positive definiteness is structural). $\hat{\mathbf{B}}^{-1}$ is positive definite by construction for any $\mathbf{r}$, because it has the form $\mathbf{L}^\top \mathbf{D} \mathbf{L}$ with $\mathbf{D}$ diagonal and positive. Truncating an already-formed precision matrix does not preserve this: in a companion measurement, masking entries beyond a given distance produced an indefinite matrix in 44 of 44 cases tested, with minimum eigenvalues as negative as -2.8×10^4 .

2 The decision variable

2.1 Clusters, not grid points

Giving every component its own radius means n parameters — here $n = 18818$ — fitted against a few hundred observations. That does not fail for want of a better search. In a companion experiment on a smaller system, an oracle with full access to the truth reduced the analysis error by 59% on the cycle it optimized, and the same radius field made the error 22% *worse* on the following cycles, performing worse than a random shuffle of its own values. The structure it found was the cycle’s noise.

We therefore group the grid points of each field into K clusters and estimate one radius per cluster. The features are computed from the forecast ensemble alone — the log background variance and the ensemble correlation at lags one and two, averaged over the four directions — so the grouping is admissible in the same sense as the criterion. Clustering is by k -means on those features.

Remark 3 (The clusters need not be spatially contiguous). *Points are grouped by how their ensemble behaves, not by where they sit. Two points at opposite corners of the domain can share a radius. One consequence matters for the search: a single decision variable controls grid points scattered over the whole domain, and the variables are close to independent of one another.*

2.2 An integer vector

With two fields, the decision variable is

$$\boldsymbol{\theta} = (\theta_1, \dots, \theta_K, \theta_{K+1}, \dots, \theta_{2K}) \in \{1, \dots, r_{\max}\}^{2K}, \quad (3)$$

the first K entries the radii of the clusters of q and the next K those of ψ . The expansion to the n per-component radii is the deterministic map $r_j = \theta_{c(j)}$, with $c(j)$ the cluster of point j ; it is not part of the search. With $K = 4$ and $r_{\max} = 12$ the space has $12^8 \approx 4.3 \times 10^8$ elements.

Nothing in the search produces a real-valued solution and nothing is rounded.

3 The objective

3.1 Why observations must be withheld

Scoring the analysis against the observations it assimilated is circular. In a direct measurement, that objective fell monotonically as the radius shrank, reaching its minimum at the smallest admissible value: a short radius makes each point fit its own observation and the residual goes to zero. The criterion rewards copying the data.

So the observation set $\mathcal{O}$ is split, $\mathcal{O} = \mathcal{T} \cup \mathcal{V}$, the analysis is computed from $\mathcal{T}$ alone, and the result is scored at $\mathcal{V}$:

$$J(\boldsymbol{\theta}) = \frac{1}{|\mathcal{V}|} \|\mathbf{y}_{\mathcal{V}} - \mathbf{H}_{\mathcal{V}} \mathbf{x}^a(\boldsymbol{\theta}; \mathcal{T})\|_{\mathbf{R}_{\mathcal{V}}^{-1}}^2. \quad (4)$$

No reference trajectory enters (4).

Algorithm 1 Tabu search on the integer radius vector

```
1:  $\theta \leftarrow$  random vector in  $\{1, \dots, r_{\max}\}^{2K}$ ;  $\theta^* \leftarrow \theta$ 
2:  $T \leftarrow \emptyset$  ▷ tabu list of (component, value) pairs
3: while budget remains do
4:    $\mathcal{C} \leftarrow \{ \theta' \in \mathcal{N}(\theta) : (j, \theta'_j) \notin T \text{ or } J(\theta') < J(\theta^*) \}$  ▷ aspiration
5:    $\theta \leftarrow \arg \min_{\theta' \in \mathcal{C}} J(\theta')$ 
6:   add  $(j, \theta_j)$  to  $T$  for  $\tau$  iterations
7:   if  $J(\theta) < J(\theta^*)$  then  $\theta^* \leftarrow \theta$ 
8:   end if
9: end while
```

3.2 One partition or many

A single partition already removes the circularity. Repeating it over B draws reduces variance, and the two natural aggregations differ:

- **Averaging** J over the B partitions and minimizing the mean.
- **Voting**: minimize each realization separately and take, for each component, the mode of the B winning values. Because the radii are integers the mode needs no binning.

Voting is the more robust of the two when the landscape is flat, since each vote only has to fall in the right place rather than locate an exact minimum. It is also the more expensive: B optimizations instead of one.

Remark 4 (A diagnostic that comes free). *The agreement among votes — how many of the B winners a component's mode collects — says whether the structure is identified or whether noise is being averaged. In the companion experiment the agreement was 7 out of 30, which is what a null result looks like.*

3.3 The oracle

Alongside (4) we run $J_{\text{oracle}}(\theta) = \|\mathbf{x}^a(\theta) - \mathbf{x}^{\text{true}}\|$, which no operational system can compute. Its purpose is to bound what any selection rule could achieve. If a clustered radius does not beat a uniform one *with the truth in hand*, there is nothing to estimate and the negative result is clean.

4 The searches

All six share one neighbourhood, so that a comparison between them measures the acceptance rule and not the move.

Definition (Neighbourhood). $\mathcal{N}(\theta)$ is the set of vectors that differ from θ in exactly one component. With a step s , the new value is drawn from $\{\theta_j - s, \dots, \theta_j + s\} \cap [1, r_{\max}]$; with $s = \infty$, from the whole range.

The two moves are the discrete reading of the continuous algorithm [4]. The Lévy step does not produce a position; it produces *how many* components a move copies, so its heavy tail gives mostly small moves with occasional large ones, which is the property the flight contributes. Following the reference implementation, the global step is taken when $\text{rand} > p$, so $p = 0.8$ means global pollination one time in five.

The remaining four:

Algorithm 2 Discrete flower pollination

```
1: initialize a population of  $S$  integer vectors;  $\theta^* \leftarrow$  best of them
2: while budget remains do
3:   for each individual  $\theta^{(s)}$  do
4:     if rand  $> p$  then ▷ global pollination
5:        $m \leftarrow$  a Lévy draw mapped to  $\{1, \dots, 2K\}$ 
6:       copy  $m$  randomly chosen components from  $\theta^*$  into  $\theta^{(s)}$ 
7:     else ▷ local pollination
8:       pick  $u, v$  at random; let  $\mathcal{D} = \{j : \theta_j^{(u)} \neq \theta_j^{(v)}\}$ 
9:       copy a random subset of  $\mathcal{D}$  from  $\theta^{(u)}$  into  $\theta^{(s)}$ 
10:    end if
11:    accept if improved; update  $\theta^*$ 
12:  end for
13: end while
```

Simulated annealing the same neighbourhood as tabu, accepting a worsening move with probability $\exp(-\Delta/T)$; the step shrinks with temperature, so it is a tabu search with an unrestricted but cooling neighbourhood rather than a different algorithm.

Genetic algorithm uniform crossover on whole components — integers have no meaningful midpoint — with one-component mutation and tournament selection.

Ant colony a pheromone table $\tau \in \mathbb{R}^{2K \times r_{\max}}$ over (component, value) pairs. Each ant builds a whole vector by choosing a value per component independently with probability $\propto \tau_{j,v}^a$; the best ants reinforce, everything evaporates. *This is the search the parameterization suits best:* because the clusters are not spatially contiguous their radii are close to independent, and the pheromone accumulates evidence about each one separately, so an ant can combine the good value of one cluster with that of another even though the two have never appeared together.

Firefly attraction becomes a probability of copying: each component of a brighter individual is copied with probability β , decaying with the Hamming distance between the two vectors.

Two **controls**, which are not competitors but the calibration of the comparison: uniform random sampling at the same budget, and an exhaustive sweep where the space is small enough to enumerate. When the sweep completes, every other search is scored as a gap to the true optimum rather than against its rivals.

5 Cost, and what it forces

5.1 Row caching

Row separability has a direct consequence. The pair (i, r_i) determines row i of $\mathbf{L}$ and entry $\mathbf{D}_{ii}$ completely, so they can be cached. A move that changes one cluster then recomputes only that cluster's rows, and a move that revisits a radius recomputes nothing.

Profiling one analysis at $n = 4802$: 7.8s in total, 7.2 in the precision, and 5.1 of those inside `scikit-learn` — 4800 calls to `Ridge.fit`, one per grid point, of which 2.1s were input validation. Each regression has four predecessors and forty members and takes microseconds to solve directly. Solving (1) in closed form and caching by row takes a warm rebuild from 3.48s to 0.126s, a factor of 28.

Remark 5 (An argument for the trajectory methods). *A move that changes one component costs a fraction of a full evaluation, while a move that changes the whole vector costs all of*

it. This is a property of the problem, not of the implementation, and it favours tabu search and annealing over the population methods in a way that budget-counted comparisons at equal evaluations do not show.

5.2 What still does not fit

At $n = 18818$ with $N = 40$, one analysis costs 11 s. Optimizing inside every assimilation cycle with B partitions and a budget of P evaluations costs $B \cdot P \cdot 11$ seconds per cycle: with $B = 10$, $P = 40$ and 40 cycles, about 49 hours per run. The options are to optimize every M cycles rather than every cycle — defensible, since the radius is supposed to vary slowly — to reduce B , or to update the analysis solve under a low-rank change instead of re-solving. The last has not been implemented.

6 The testbed

The model is the 1.5-layer quasi-geostrophic system of Sakov and Oke [3],

$$q_t = -\psi_x - \varepsilon J(\psi, q) - A \Delta^3 \psi + 2\pi \sin(2\pi y), \quad q = \Delta \psi - F\psi, \quad (5)$$

on the unit square with $\psi = \Delta \psi = \Delta^2 \psi = 0$ on the boundary, $F = 1600$, $\varepsilon = 10^{-5}$. It was chosen because Lorenz-96 cannot answer the question: with a uniform grid, uniform forcing and identical observations everywhere, no grid point has any reason to want a different radius from its neighbour, so a negative result there says nothing about the idea. The QG model has two fields with genuinely different correlation scales — q carries fine filaments while ψ solves a Helmholtz problem on q and is smooth — and a flow that is not homogeneous.

Three settings are consequences of measurements rather than choices.

The dissipation is ten times the model default. At $\text{rkh2} = 10^{-12}$ the norm of q grows without saturating, a factor of fourteen between $t = 500$ and $t = 8000$, so there is no stationary regime and no climatology to sample. At 10^{-11} it settles: 1.107×10^4 at $t = 20000$ against 1.222×10^4 at $t = 60000$. Sakov and Oke report the same, raising the dissipation by ten and reducing the step from 1.5 to 1.25 for a stable assimilating system.

The ensemble is climatological. One long run, snapshots along it, members and truth drawn at random from that set. Perturbing a state and propagating does not work here, and the reason was measured: a perturbation is mostly energy off the attractor and the hyperviscosity removes it. Starting from 30% of climatology the background error falls to 0.068 after 120 time units, a factor of five, and raising the amplitude does not compensate because the propagation eats it. Only after about 250 units does the surviving component grow, reaching 0.872 at $t = 1000$ — exactly what a climatological ensemble has from the start.

Observations sit on a regular lattice. With a random subset the spacing fluctuates, so some local domains hold several observations and others none, and the analysis at a point with an empty domain is just the background. Measured at 2% random density, the analysis left q unchanged to four decimal places at *every* radius, which looks like a broken filter and is not one.

7 The experiments

EXP-01 sweeps the observation lattice spacing against a uniform radius and reports, per density, whether the optimum falls in the *interior* of the admissible range. An optimum at the boundary means the configuration has not reached the regime where the radius trades one error against another, and no estimation experiment run there would mean anything.

Identifier	Question	Output
EXP-00-SETUP	Is the testbed what it claims to be?	climatology, ensemble, lattice
EXP-01-REGIME	Where is the radius a two-sided decision?	optimum vs. observation density
EXP-02-CLUSTERS	Does a clustered radius pay, and which search finds it?	gain over uniform; search compa

EXP-02 runs the six searches and the two controls at equal budget, under both (4) and the oracle, for $K \in \{1, 2, 3, 4, 6\}$.

8 Open problems

The precision matrices do not yet agree to machine precision. On a single row the closed-form ridge reproduces `scikit-learn` to 2.4×10^{-16} . After the boundary components were excluded from estimation — q vanishes there, so $1/\text{var}$ is infinite, and `pyteda` produces 290 infinities at $n = 4802$ — the assembled matrices differ by tens of percent, with the largest discrepancies on diagonal entries of order 10^{13} . Whether that is confined to those degenerate rows or whether excluding the boundary changes the interior regressions has not been traced. The equivalence test is marked expected-to-fail rather than relaxed until it passes.

Simulated annealing underperforms random sampling. On a separable quadratic over four components, averaged across six seeds, tabu reaches the exact optimum every time, random sampling reaches 3.0, and annealing reaches 4.0; slower cooling makes it worse. It spends too much of a small budget on temperature levels. It is left in the comparison as it performs.

The state is normalized per field, and that is an implementation decision. Each field is divided by its climatological spread before the analysis and restored after. The reason is the observation error: `pyteda`’s LETKF reads a single scalar variance, so with q of order 10^3 and ψ of order 1 no isotropic error suits both. It is not part of the modified-Cholesky method and can be removed by passing a heterogeneous $\mathbf{R}$.

References

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